

Two methods for inference

Math 213

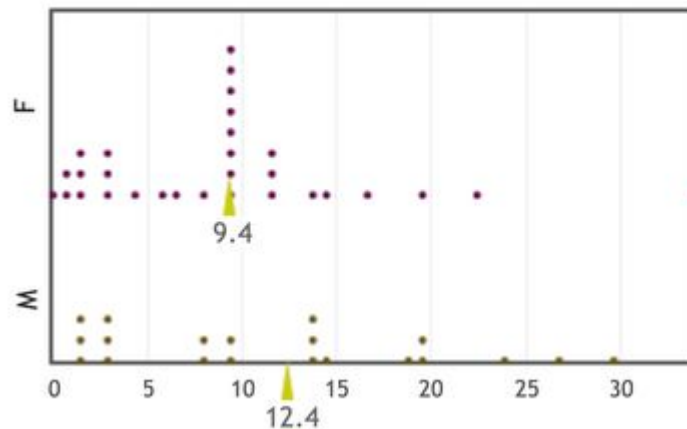
Exercise Hours by Gender

A sample of 50 people (20 men, 30 women) were asked the number of hours a week they exercised. Here are the dotplots for this variable.

We see that, on average, the men exercise 3 hours more per week than the women. But can this amount of variation just be due to chance? Or are the two groups really different?

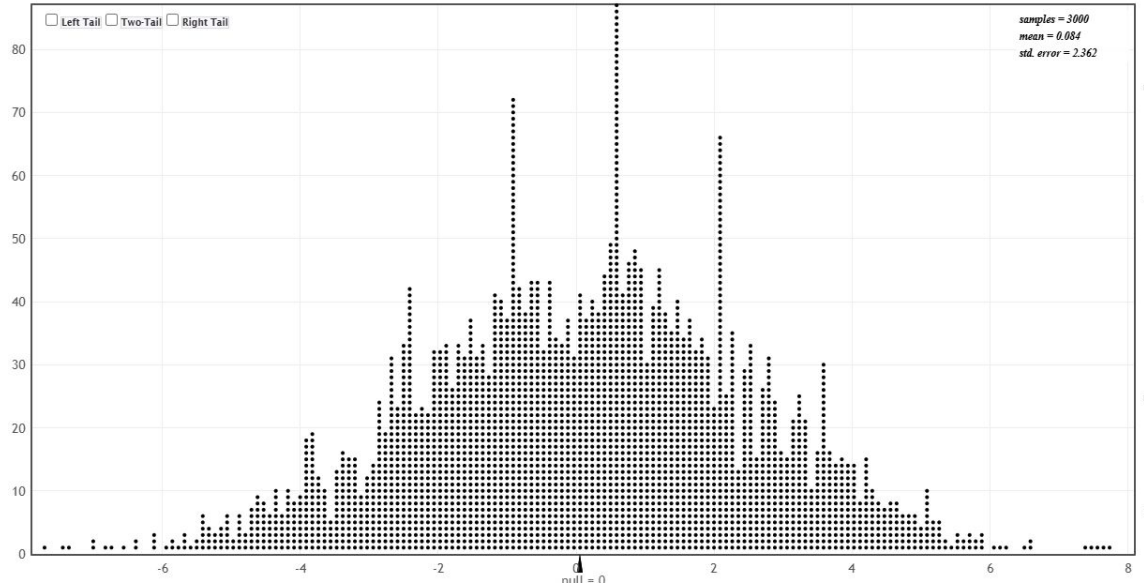
Original Sample

$$\bar{x}_1 - \bar{x}_2 = 3, n_1 = 20, n_2 = 30$$



Group 1 - Randomization Tests

Randomization tests - In Statkey, select the Randomization Test for Difference in Means tool and select “Exercise Hours (Male vs Female).” Generate the randomization test dotplot with a few thousand samples and paste it here.



1. Briefly explain how that dotplot is made. For each dot, what is being calculated? Remember what we did with index cards and the weight loss study.

This dotplot was made by taking the responses and shuffling them between groups. The difference between the randomized groups is plotted. This is repeated 2999 more times.

2. What are the null and alternate hypotheses? Use the appropriate symbols.

$$H_0: \mu_m - \mu_f = 0$$

$$H_a: \mu_m - \mu_f > 0$$

3. What is the major assumption that underlies this plot?

The null hypothesis is true

4. Where is a randomization test dotplot always centered? Write a sentence explaining why it's centered there.

It's centered around the null hypothesis, because it's made assuming the null is true.

5. In general, how do you use a sample statistic and the dotplot to draw a conclusion about the difference between the two groups?

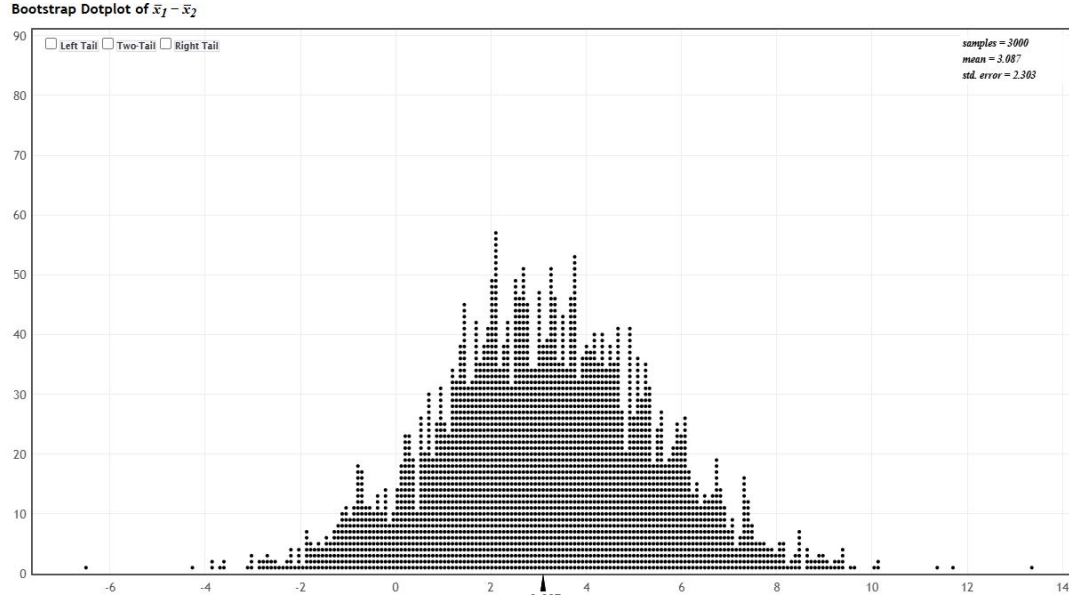
You consider what percentage of the data in the dotplot is as or more extreme as the sample statistic. If the percentage is <0.05 , you can reasonably reject the null hypothesis

6. What conclusion can you draw from our sample statistic and dotplot? You can use the p-value calculating tool if that would help.

The p-value is 0.108, which means we do not have enough evidence to reject the null hypothesis

Bootstrap sampling distribution

Now select the Confidence Interval for a Difference in Means tool in statkey. Again find the exercise hours dataset and generate a dotplot.



1. Briefly explain how *this* dotplot is made. For each dot, what is being calculated? How is “bootstrapping” involved?

Each dot is the difference in means for a bootstrapped sample. Bootstrapping involves taking samples from the data we already have, with replacement. This means that the same value can be added to the bootstrapped sample more than once.

Create a simulated population by making many, many copies of our original data and then taking samples (of 20 men and 30 women) from the “bootstrapped” population. Calculate $\bar{x}_m - \bar{x}_f$ and plot that.

2. Where are *these* dotplots always centered? Write a sentence explaining why it makes sense that they are centered there.

They are centered near the sample statistic. Maybe because the sample is the “population”

3. In general, how do you use this dotplot to draw conclusions about a population parameter (or two population parameters, in this case).

We can use the quantiles to determine how confident we are in the difference between the 2 population parameters.

4. Follow your procedure to generate the appropriate inference for these variables. What can you conclude? Write your explanation the way we've practiced.

We are 95% confident that the average man truly exercises between 1.3 hours less than and 7.6 hours more per week than the average woman.

We are 95% confident that the difference in avg weekly exercise hours between men and women is between -1.4 and 8 hours.

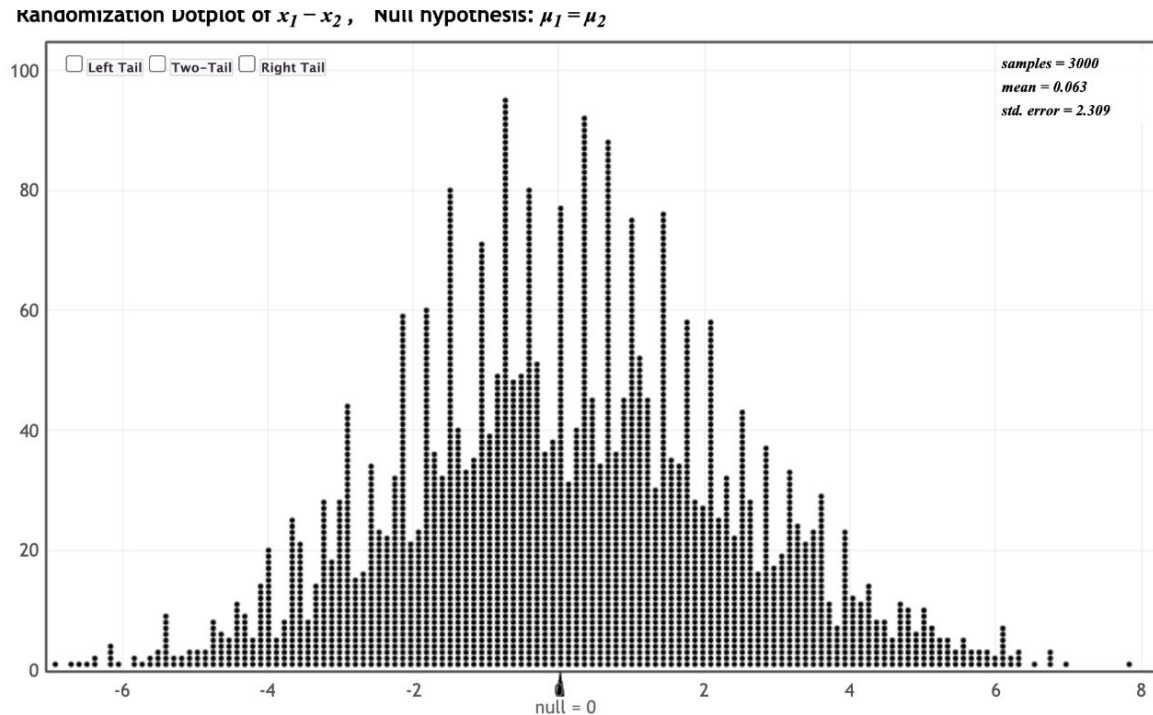
5. What does "95% confidence" mean?

This method tends to work 95% of the time

By design, 95% of the time we make a confidence interval, it will contain the true pop param

Group 2 - Randomization Tests

Randomization tests - In Statkey, select the Randomization Test for Difference in Means tool and select “Exercise Hours (Male vs Female).” Generate the randomization test dotplot with a few thousand samples and paste it here.



1. Briefly explain how that dotplot is made. For each dot, what is being calculated? Remember what we did with index cards and the weight loss study.

The dot plot was made by taking random samples of exercise hours, and then finding the means based on the exercise hours of females and males respectively, which are then subtracted by each other. This process is then repeated around 3000 more times. Each dot represents the difference in the mean exercise hours of females and males per sample

2. What are the null and alternate hypotheses? Use the appropriate symbols.

$H_a: \mu_m - \mu_f > 0$

$H_n: \mu_m - \mu_f = 0$

3. What is the major assumption that underlies this plot?

That the null hypothesis is true

4. Where is a randomization test dotplot always centered? Write a sentence explaining why it's centered there.

It's always centered around the null hypothesis because we are assuming the null hypothesis is true.

5. In general, how do you use a sample statistic and the dotplot to draw a conclusion about the difference between the two groups?

The dotplot is meant to represent the range of data that would be expected if the null hypothesis was true. Therefore, we can then see if the result we got for the sample statistic can comfortably be placed amongst that random chance, or if it's unlikely that point would have been gotten in that fashion.

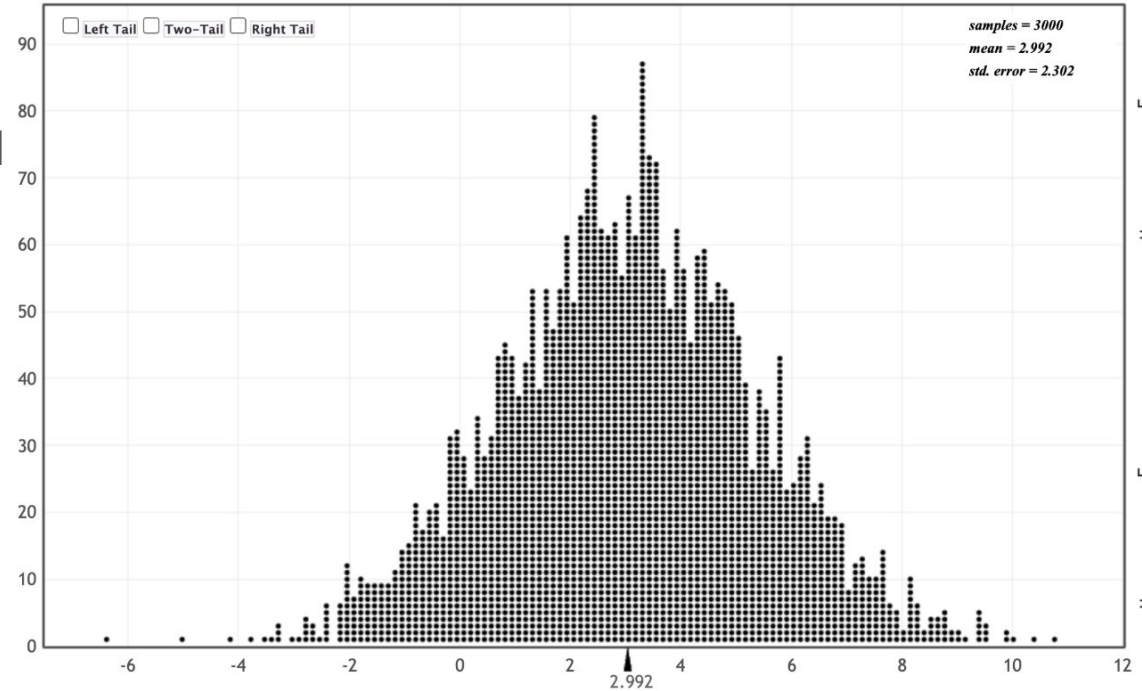
6. What conclusion can you draw from our sample statistic and dotplot? You can use the p-value calculating tool if that would help.

From the dotplot provided, it seems as though a 3 hour difference fits comfortably amongst the randomization graph's 95% proportion. Therefore, it is unlikely that this statistic rejects the null hypothesis

Bootstrap sampling distribution

Now select the Confidence Interval for a Difference in Means tool in statkey. Again find the exercise hours dataset and generate a dotplot.

Bootstrap Dotplot of $\bar{x}_1 - \bar{x}_2$



1. Briefly explain how *this* dotplot is made. For each dot, what is being calculated? How is “bootstrapping” involved?

This dotplot is made by each sample randomly being generated from what was already given in the experiment, like picking chips from a cup, recording them, and then putting them back. This creates samples from actual data points, but collected at random with a chance of repeat. Therefore, each dot represents the difference between the means of exercise hours between males and females per each of these samples.

2. Where are *these* dotplots always centered? Write a sentence explaining why it makes sense that they are centered there.

It's normally always centered around our observed sample statistic. This makes sense because the data points that are given for each sample are drawn from the original, not just at random, so it's more likely that the sample will often be reflective of the sample statistic.

3. In general, how do you use this dotplot to draw conclusions about a population parameter (or two population parameters, in this case).

We can use the quantiles in order to create confidence intervals of the range of the difference in exercise hours for a percentage of the population.

4. Follow your procedure to generate the appropriate inference for these variables. What can you conclude? Write your explanation the way we've practiced.

We are 95% confident that men tend to exercise more than female by a range of -1.583 to 7.533 hours a week.

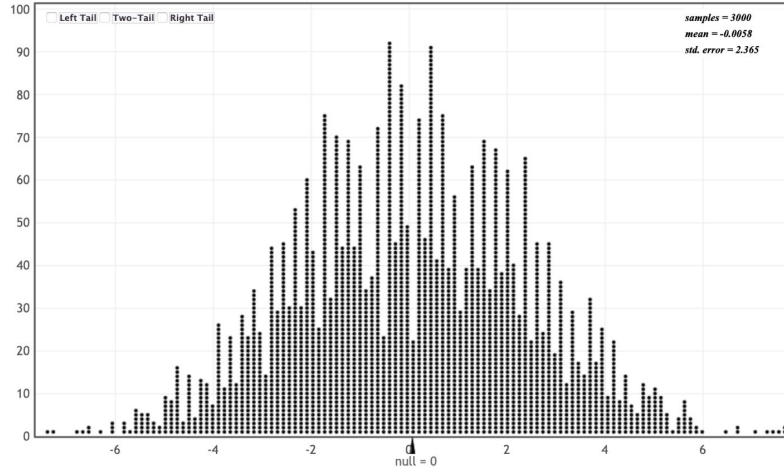
5. What does "95% confidence" mean?

It means that this range is suspected to include 95% of the population we are trying to generalize to.

Group 3 - Randomization Tests

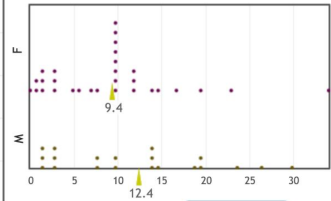
Randomization tests - In Statkey, select the Randomization Test for Difference in Means tool and select “Exercise Hours (Male vs Female).” Generate the randomization test dotplot with a few thousand samples and paste it here.

Randomization Dotplot of $\bar{x}_1 - \bar{x}_2$, Null hypothesis: $\mu_1 = \mu_2$



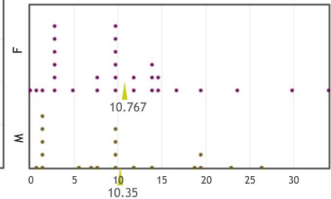
Original Sample

$\bar{x}_1 - \bar{x}_2 = 3$, $n_1 = 20$, $n_2 = 30$



Randomization Sample

$\bar{x}_1 - \bar{x}_2 = -0.42$, $n_1 = 20$, $n_2 = 30$



1. Briefly explain how that dotplot is made. For each dot, what is being calculated? Remember what we did with index cards and the weight loss study.

Each plot shows a difference in mean, assuming the null is true.

2. What are the null and alternate hypotheses? Use the appropriate symbols.

$$H_0: \mu_m - \mu_f = 0$$

$$H_1: \mu_m - \mu_f > 0$$

3. What is the major assumption that underlies this plot?

The null is true, there is no difference between exercise hours between women and men.

4. Where is a randomization test dotplot always centered? Write a sentence explaining why it's centered there.

Always centered at 0, since it assumes there is no difference.

5. In general, how do you use a sample statistic and the dotplot to draw a conclusion about the difference between the two groups?

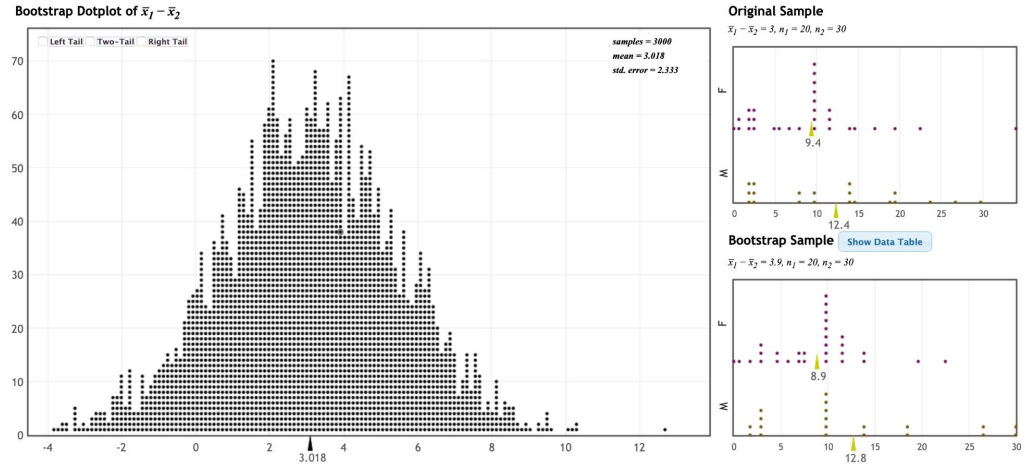
You compare your sample distribution to the randomization group, if the observed difference is far in the tail, or not in the plot at all, you have reason to reject the null hypothesis

6. What conclusion can you draw from our sample statistic and dotplot? You can use the p-value calculating tool if that would help.

The observed difference was 3, while the randomization group was centered at 0. Since this difference was seen multiple times we can fail to reject the null.

Bootstrap sampling distribution

Now select the Confidence Interval for a Difference in Means tool in statkey. Again find the exercise hours dataset and generate a dotplot.



1. Briefly explain how *this* dotplot is made. For each dot, what is being calculated? How is “bootstrapping” involved?

Each plot represents a difference of mean from one bootstrap sample. We resample with replacement from each group multiple times, calculate the difference and then plot them.

2. Where are *these* dotplots always centered? Write a sentence explaining why it makes sense that they are centered there.

Is always centered around the observed statistic, since all resamples are taken from the original sample.

3. In general, how do you use this dotplot to draw conclusions about a population parameter (or two population parameters, in this case).

We use the spread of the dotplot to build confidence intervals, if the interval does not contain 0, we have evidence for meaningful difference between the population means. This interval estimates the likely range for the true difference.

4. Follow your procedure to generate the appropriate inference for these variables. What can you conclude? Write your explanation the way we've practiced.

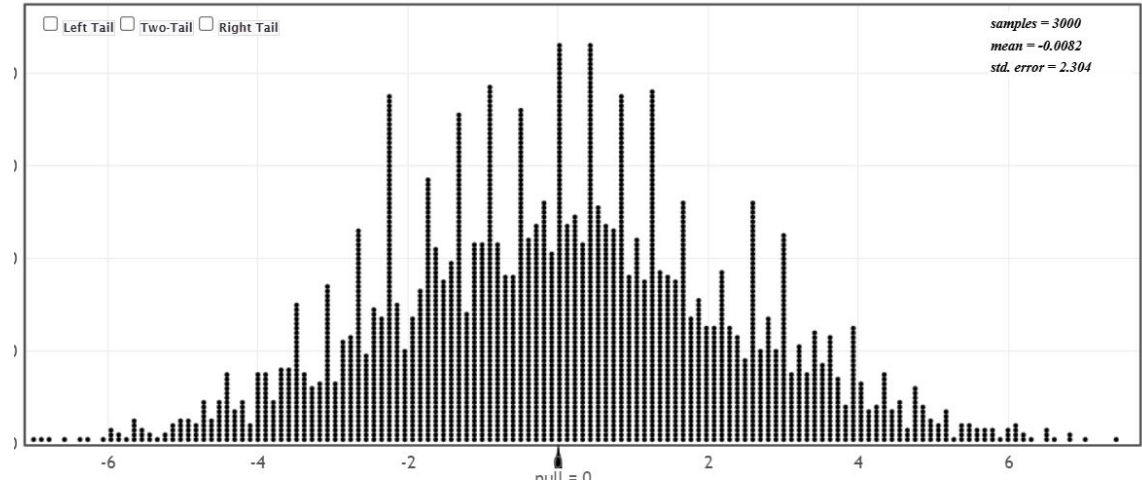
Based on our bootstrap distribution and the original sample difference of 3, we are 95% confident that the true difference in mean exercise hours between females and males is between approximately 0.6 and 5.4 hours.

5. What does "95% confidence" mean?

means that 95% of the time, the method to produce a 95% confidence interval will create a range that includes the population parameter.

Group 4 - Randomization Tests

Randomization tests - In Statkey, select the Randomization Test for Difference in Means tool and select “Exercise Hours (Male vs Female).” Generate the randomization test dotplot with a few thousand samples and paste it here.



1. Briefly explain how that dotplot is made. For each dot, what is being calculated? Remember what we did with index cards and the weight loss study.
 - a. Sampling with replacement from given sample, like writing all average hours on index cards, shuffling, sorting 20 to male and 30 to female, calculate difference of means of these piles
 - b. Each dot is a simulated difference in the mean number of hours exercised by men vs women for a sample of 20 men vs 30 women.
2. What are the null and alternate hypotheses? Use the appropriate symbols.
 - a. $\mu_m - \mu_f = 0$; There is no difference between men and women for the average amount of hours they exercise per week.
 - b. $\mu_m - \mu_f > 0$: There is a difference in means between the number of hours men and women exercise and it's not due to random chance.
3. What is the major assumption that underlies this plot?
 - a. That the null hypothesis is true

4. Where is a randomization test dotplot always centered? Write a sentence explaining why it's centered there.

It's centred at 0 because that is our null hypothesis.

5. In general, how do you use a sample statistic and the dotplot to draw a conclusion about the difference between the two groups?

See how frequently we have results as extreme or more extreme than the sample statistic.

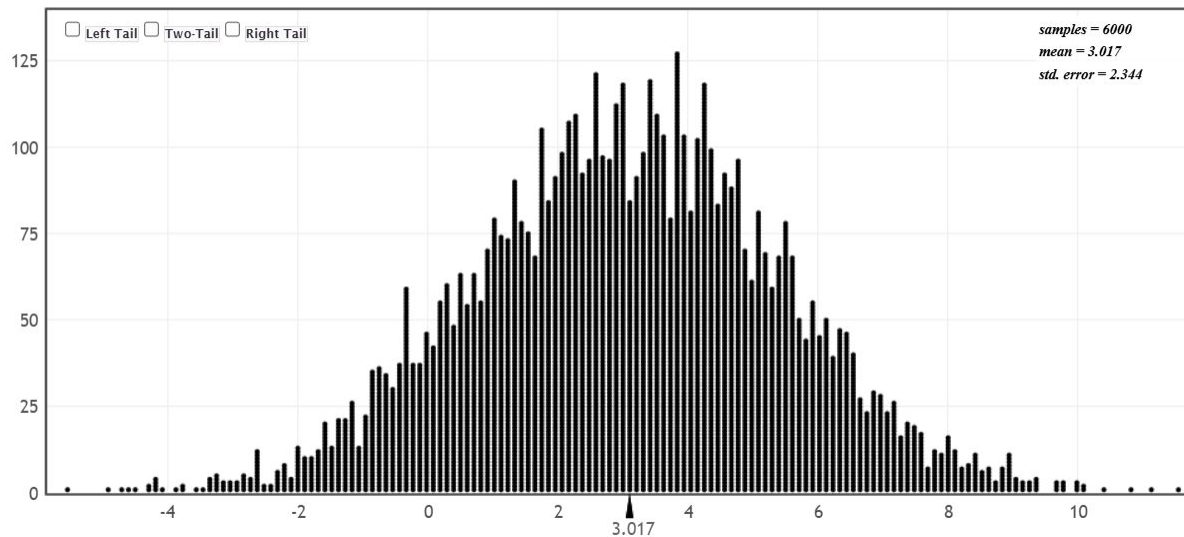
6. What conclusion can you draw from our sample statistic and dotplot? You can use the p-value calculating tool if that would help.

$p=.105$ - We fail to reject the null hypothesis as the difference in means falls within 95% of the simulated difference of means.

Bootstrap sampling distribution

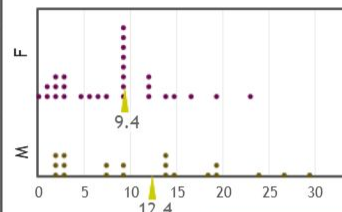
Now select the Confidence Interval for a Difference in Means tool in statkey. Again find the exercise hours dataset and generate a dotplot.

Bootstrap Dotplot of $\bar{x}_1 - \bar{x}_2$



Original Sample

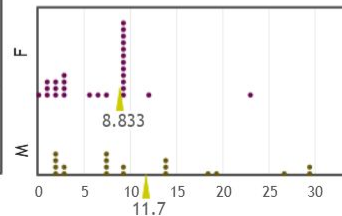
$\bar{x}_1 - \bar{x}_2 = 3$, $n_1 = 20$, $n_2 = 30$



Bootstrap Sample

[Show Data Table](#)

$\bar{x}_1 - \bar{x}_2 = 2.87$, $n_1 = 20$, $n_2 = 30$



1. Briefly explain how *this* dotplot is made. For each dot, what is being calculated? How is “bootstrapping” involved
 - a. Bootstrapping creates new samples by using sampling with replacement and allowing duplicates of the samples to be used. So each dot is a simulated difference in means for a simulated sample set bootstrapped from the original sample set to create a new simulated difference in the means.

2. Where are *these* dotplots always centered? Write a sentence explaining why it makes sense that they are centered there.

They are always centered at the sample statistic, which makes sense because the samples of differences are being built from the original sample

3. In general, how do you use this dotplot to draw conclusions about a population parameter (or two population parameters, in this case).

Calculate the standard error for the difference in means and then use the standard error to give an interval based on the sample statistic (which is the simulated mean). This will give us a confidence interval of 95% (or whatever percent we used to gather our standard error) and ensure that our population parameter, which we do not actually know at this time, is within our interval.

4. Follow your procedure to generate the appropriate inference for these variables. What can you conclude? Write your explanation the way we've practiced.

Using the p-value generator on StatKey gives us a confidence interval of (-1.525, 7.575).

$$3.017 - (2 * SE) = 3.017 - (2 * 2.344) = -1.671$$

$$3.017 + (2 * SE) = 3.017 + (2 * 2.344) = 7.705$$

Using our procedure we get a CI of (-1.671, 7.705).

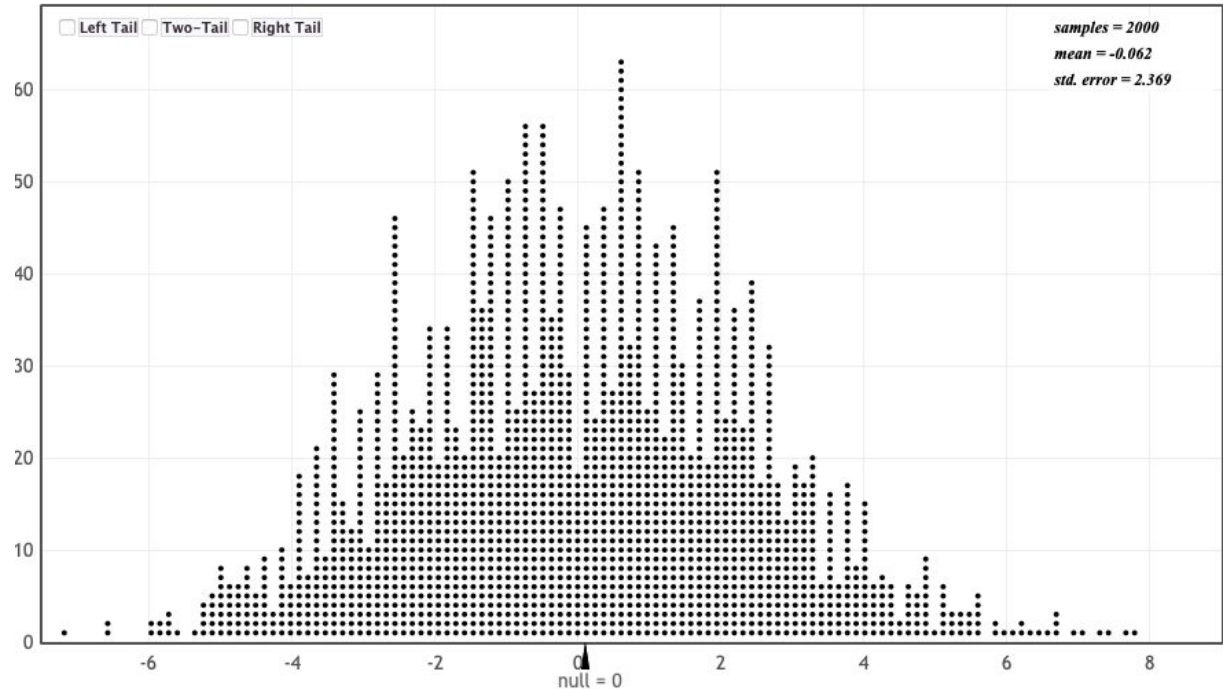
5. What does “95% confidence” mean?

95% confidence means that, in 95% of the simulations, the true population parameter will be between -1.525 and 7.575 hours.

Group 5 - Randomization Tests

Randomization tests - In Statkey, select the Randomization Test for Difference in Means tool and select “Exercise Hours (Male vs Female).” Generate the randomization test dotplot with a few thousand samples and paste it here.

Randomization Dotplot of $\bar{x}_1 - \bar{x}_2$, Null hypothesis: $\mu_1 = \mu_2$



1. Briefly explain how that dotplot is made. For each dot, what is being calculated? Remember what we did with index cards and the weight loss study.

The dotplot is the result of the difference of means for the average hours exercised by each gender for each simulated sample.

2. What are the null and alternative hypotheses? Use the appropriate symbols.

$H_0: u_m - u_f = 0$, there is no difference between the mean number of hours each gender exercises.

$H_a: u_m - u_f > 0$, there is a difference

3. What is the major assumption that underlies this plot?

The null hypothesis is true.

4. Where is a randomization test dotplot always centered? Write a sentence explaining why it's centered there.

At the value of the null hypothesis.

5. In general, how do you use a sample statistic and the dotplot to draw a conclusion about the difference between the two groups?

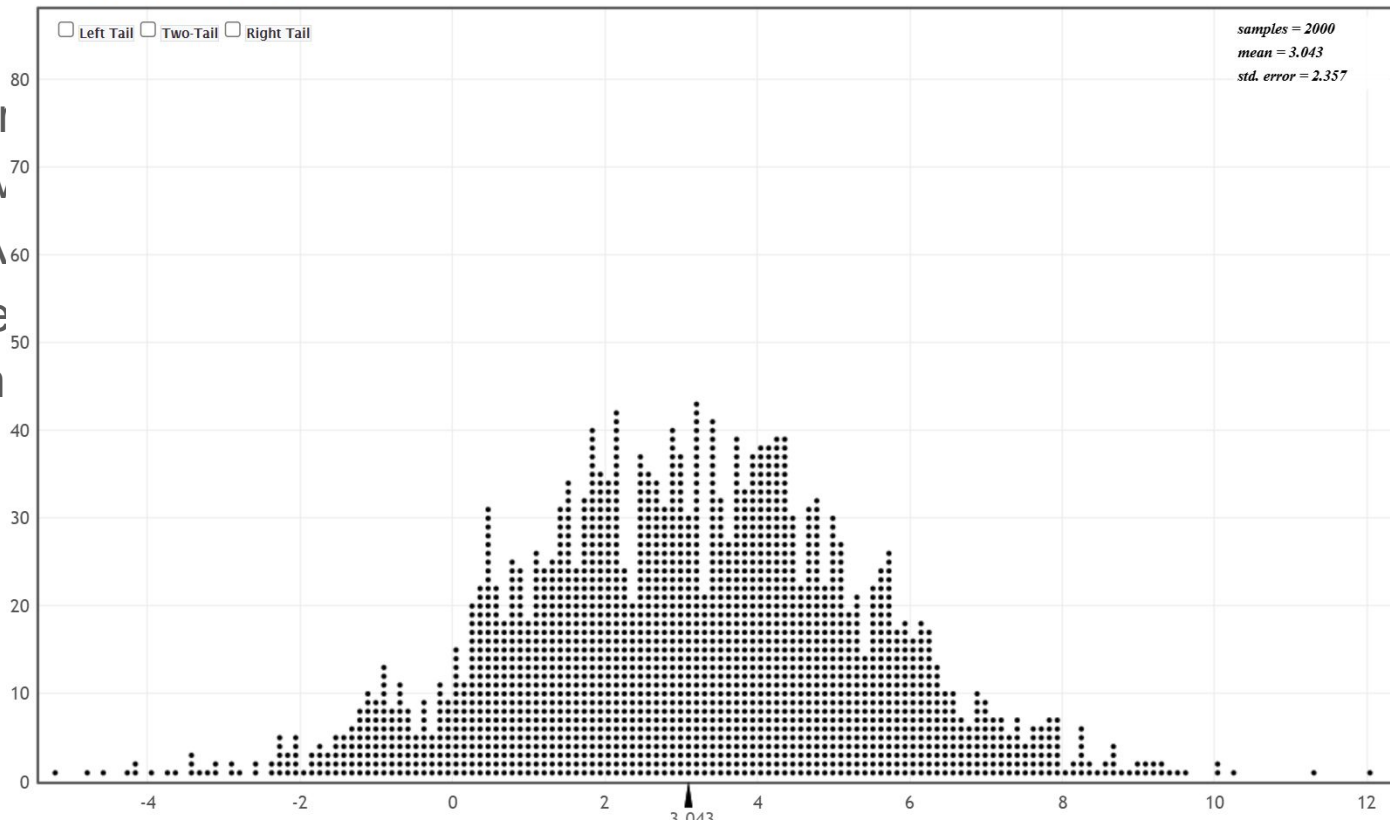
To see if any value is too extreme in comparison to the sample statistic. Malec here - not true! Also you're getting this mixed up with confidence intervals in the next answer.

6. What conclusion can you draw from our sample statistic and dotplot? You can use the p-value calculating tool if that would help.

Therefore we fail to reject the null as the value falls within 95% of the difference of means.

Bootstrap sampling distribution

Bootstrap Dotplot of $\bar{x}_1 - \bar{x}_2$



Now select the Confidence Interval for a Difference in Means tool in statkey. A pop-up will appear to find the exercise dataset and generate the dotplot.

1. Briefly explain how *this* dotplot is made. For each dot, what is being calculated? How is “bootstrapping” involved?

Bootstrapping by resampling the original data set to create simulated samples, each dot represents the difference in means from the selected “male” and “female”. The dotplot is made by taking the difference in means from these “bootstrapped” population.

2. Where are *these* dotplots always centered? Write a sentence explaining why it makes sense that they are centered there.

They are centered at the sample statistic since the simulated differences are compared to the sample statistic.

3. In general, how do you use this dotplot to draw conclusions about a population parameter (or two population parameters, in this case).

We use the 95% confidence interval to see how confident we are in the population parameter, and if the value is reasonable (within the interval). **Be more specific!**

4. Follow your procedure to generate the appropriate inference for these variables. What can you conclude? Write your explanation the way we've practiced.

We are 95% confident that difference in average weekly exercise hours between men and women is between -1.5 and 7.5 hours.

5. What does "95% confidence" mean?

The difference in means can range anywhere between -1.5 and 7.5 hours, and no difference (0 hours) is a plausible value.

Group 6 - Randomization Tests

Randomization tests - In Statkey, select the Randomization Test for Difference in Means tool and select “Exercise Hours (Male vs Female).” Generate the randomization test dotplot with a few thousand samples and paste it here.

1. Briefly explain how that dotplot is made. For each dot, what is being calculated? Remember what we did with index cards and the weight loss study.
2. What are the null and alternate hypotheses? Use the appropriate symbols.
3. What is the major assumption that underlies this plot?

4. Where is a randomization test dotplot always centered? Write a sentence explaining why it's centered there.

5. In general, how do you use a sample statistic and the dotplot to draw a conclusion about the difference between the two groups?

6. What conclusion can you draw from our sample statistic and dotplot? You can use the p-value calculating tool if that would help.

Bootstrap sampling distribution

Now select the Confidence Interval for a Difference in Means tool in statkey. Again find the exercise hours dataset and generate a dotplot.

1. Briefly explain how *this* dotplot is made. For each dot, what is being calculated? How is “bootstrapping” involved?

2. Where are *these* dotplots always centered? Write a sentence explaining why it makes sense that they are centered there.

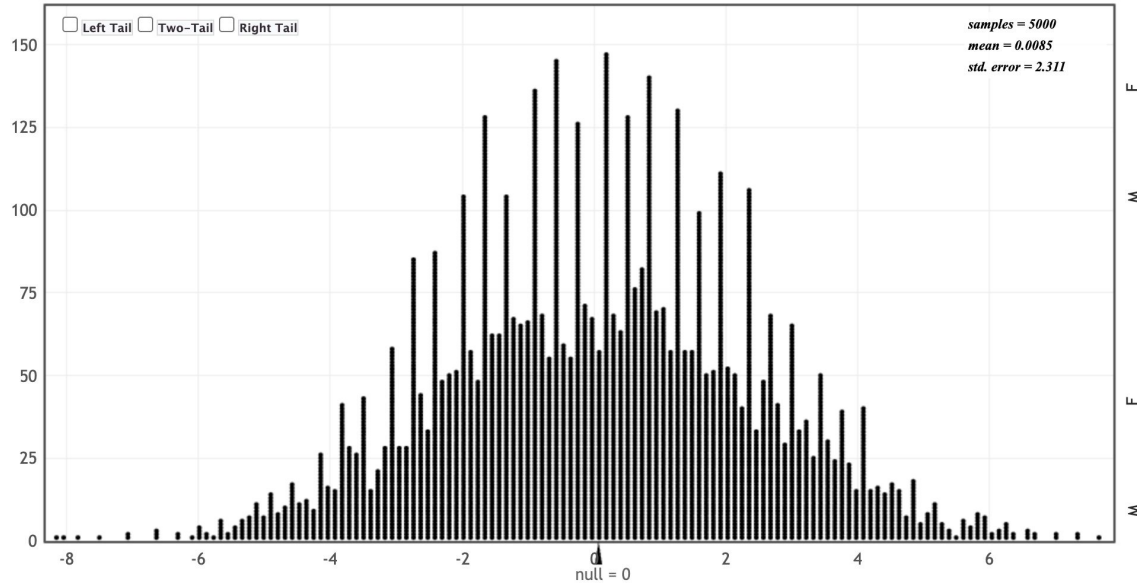
3. In general, how do you use this dotplot to draw conclusions about a population parameter (or two population parameters, in this case).

4. Follow your procedure to generate the appropriate inference for these variables. What can you conclude? Write your explanation the way we've practiced.

5. What does "95% confidence" mean?

Group 7 - Randomization Tests

Randomization tests - In Statkey, select the Randomization Test for Difference in Means tool and select “Exercise Hours (Male vs Female).” Generate the randomization test dotplot with a few thousand samples and paste it here.



1. Briefly explain how that dotplot is made. For each dot, what is being calculated? Remember what we did with index cards and the weight loss study.

Each dot is a sample of 50 people, 30 females, 20 men, and the difference between the hours worked out by men and females are plotted

2. What are the null and alternate hypotheses? Use the appropriate symbols.

H0: The difference in means of hours exercised between men and women is 0

$$\mu_{\text{diff}} = 0$$

H1: The difference in means of hours exercised between men and women is greater than 0

$$\mu_{\text{diff}} > 0$$

3. What is the major assumption that underlies this plot?

4. Where is a randomization test dotplot always centered? Write a sentence explaining why it's centered there.

Its centered assuming the null hypothesis is true (difference is 0)

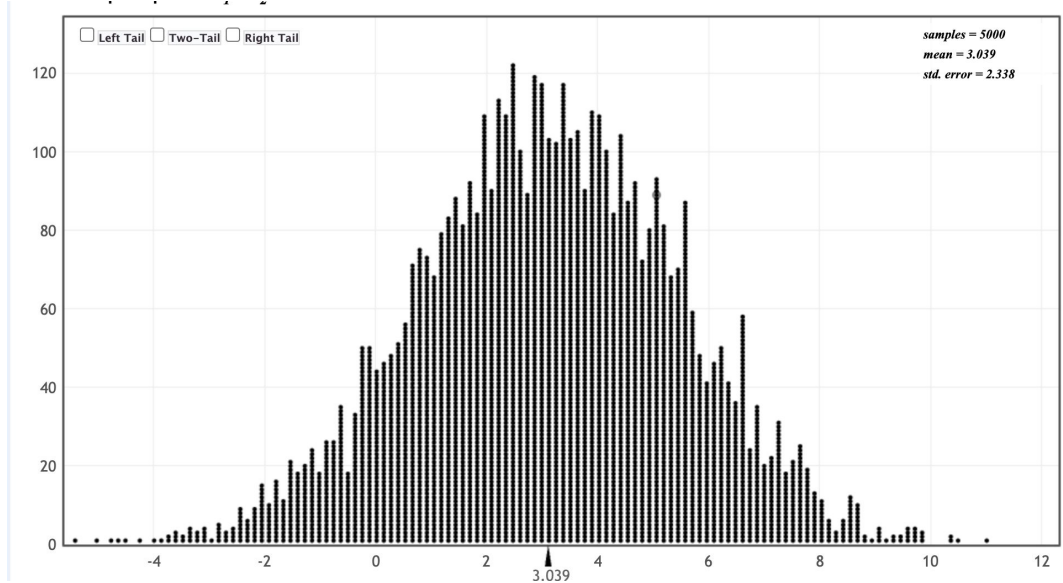
5. In general, how do you use a sample statistic and the dotplot to draw a conclusion about the difference between the two groups?

If there are many samples with a difference in sample means greater than our difference, then the chance it occurred to chance is high, which means we can't draw conclusions

6. What conclusion can you draw from our sample statistic and dotplot? You can use the p-value calculating tool if that would help.

Bootstrap sampling distribution

Now select the Confidence Interval for a Difference in Means tool in statkey. Again find the exercise hours dataset and generate a dotplot.



1. Briefly explain how *this* dotplot is made. For each dot, what is being calculated? How is “bootstrapping” involved?

Each dot is an artificial sample from our original sample

2. Where are *these* dotplots always centered? Write a sentence explaining why it makes sense that they are centered there.

These dotplots are centered at the difference in means

3. In general, how do you use this dotplot to draw conclusions about a population parameter (or two population parameters, in this case).

You find a 95% confidence interval and see if your original sample is within those intervals. If it is, then you're 95% confident that the true population parameter is within that 95% interval **The original sample is smack in the middle by design!**

4. Follow your procedure to generate the appropriate inference for these variables. What can you conclude? Write your explanation the way we've practiced.

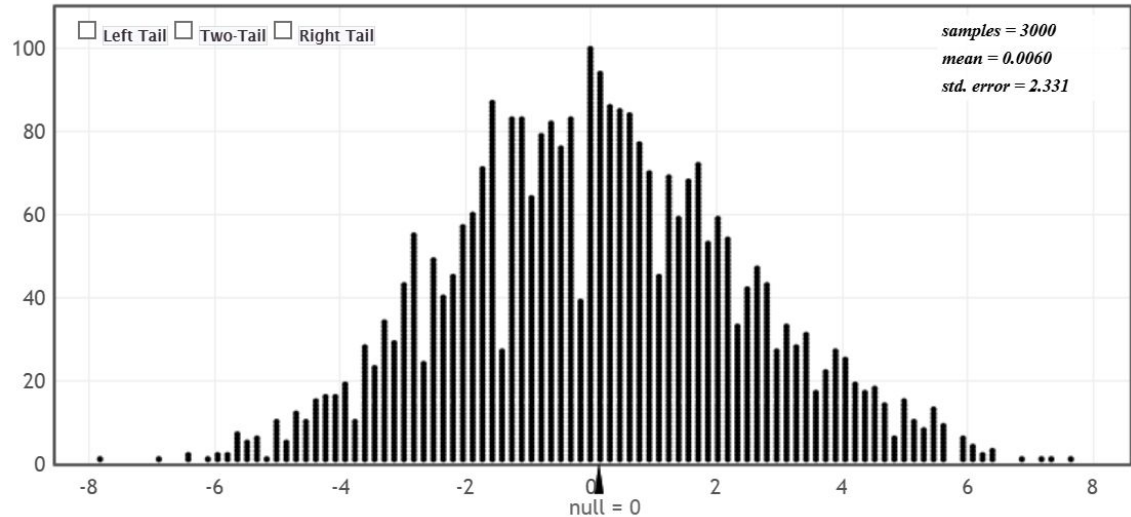
We are 95% confident that the true difference between the hours worked out by men and women is between $(3 - 2(2.338))$ and $(3 + 2(2.338))$

5. What does “95% confidence” mean?

95 times out of a 100, our bootstrap sample will contain the true population parameter

Group 8 - Randomization Tests

Randomization tests - In Statkey, select the Randomization Test for Difference in Means tool and select “Exercise Hours (Male vs Female).” Generate the randomization test dotplot with a few thousand samples and paste it here.



1. Briefly explain how that dotplot is made. For each dot, what is being calculated? Remember what we did with index cards and the weight loss study.

Each dot represents the difference in average exercise hours between the male and female groups of a random sample with replacement, under the assumption of the null hypothesis (that for the observed values, the group does not matter)

2. What are the null and alternate hypotheses? Use the appropriate symbols.

$$H_0: \mu_{\text{male}} - \mu_{\text{female}} = 0$$

$$H_A: \mu_{\text{male}} - \mu_{\text{female}} > 0$$

3. What is the major assumption that underlies this plot?

That the null hypothesis is true (nothing interesting is happening)

4. Where is a randomization test dotplot always centered? Write a sentence explaining why it's centered there.

A randomization test dotplot is always centered around the value where the null hypothesis is true (in this case it's difference between means, so it is centered around 0)

5. In general, how do you use a sample statistic and the dotplot to draw a conclusion about the difference between the two groups?

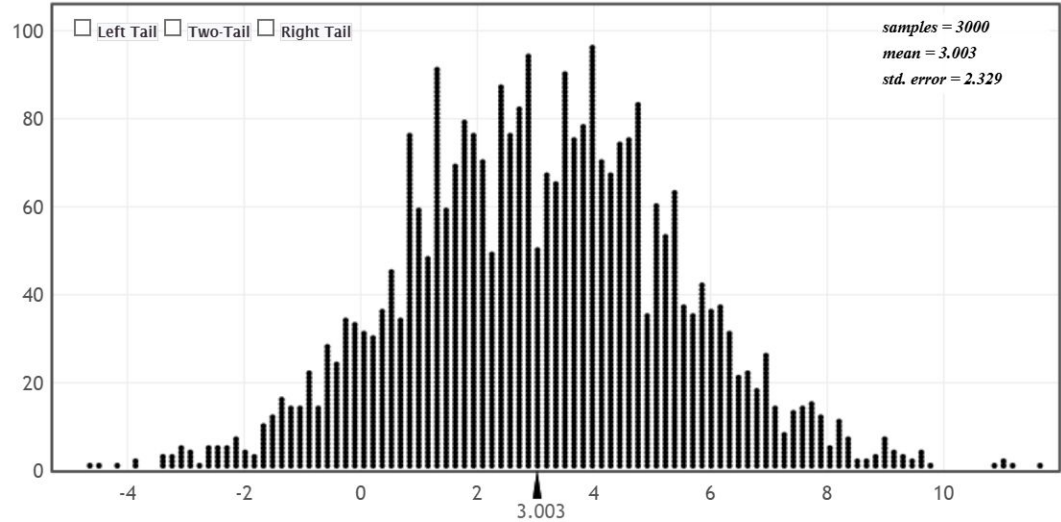
If the sample statistic value on the randomization test dotplot is unlikely (i.e., not many dots $\geq$ the sample statistic), then we reject the null hypothesis. Otherwise, we fail to reject the null hypothesis.

6. What conclusion can you draw from our sample statistic and dotplot? You can use the p-value calculating tool if that would help.

Sample statistic = 3. The p-value is 0.110. If our standard is 0.05, then we fail to reject the null hypothesis.

Bootstrap sampling distribution

Now select the Confidence Interval for a Difference in Means tool in statkey. Again find the exercise hours dataset and generate a dotplot.



1. Briefly explain how *this* dotplot is made. For each dot, what is being calculated? How is “bootstrapping” involved?

Each dot represents the difference between average exercise hours between the male and female groups of a bootstrapped sample. A bootstrapped sample takes a random sample with replacement from the original sample, per group. This becomes a simulated population.

2. Where are *these* dotplots always centered? Write a sentence explaining why it makes sense that they are centered there.

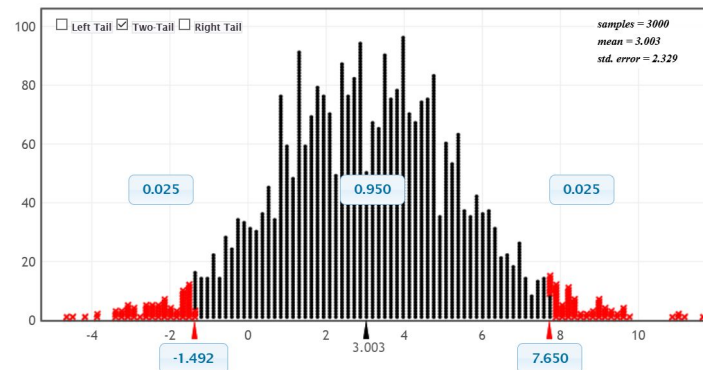
These dotplots are always centered at the original sample statistic. This makes sense because we are taking a randomized sample of it to create a simulated population and calculate the difference in means between the two groups (representing a dot). This results in a bell shaped dotplot around the original sample statistic.

3. In general, how do you use this dotplot to draw conclusions about a population parameter (or two population parameters, in this case).

We can calculate a confidence interval to represent our confidence that the population parameter will be within this interval.

4. Follow your procedure to generate the appropriate inference for these variables. What can you conclude? Write your explanation the way we've practiced.

We can conclude that we are 95% confident that the difference in average exercise hours between males and females is between -1.492 and 7.650.

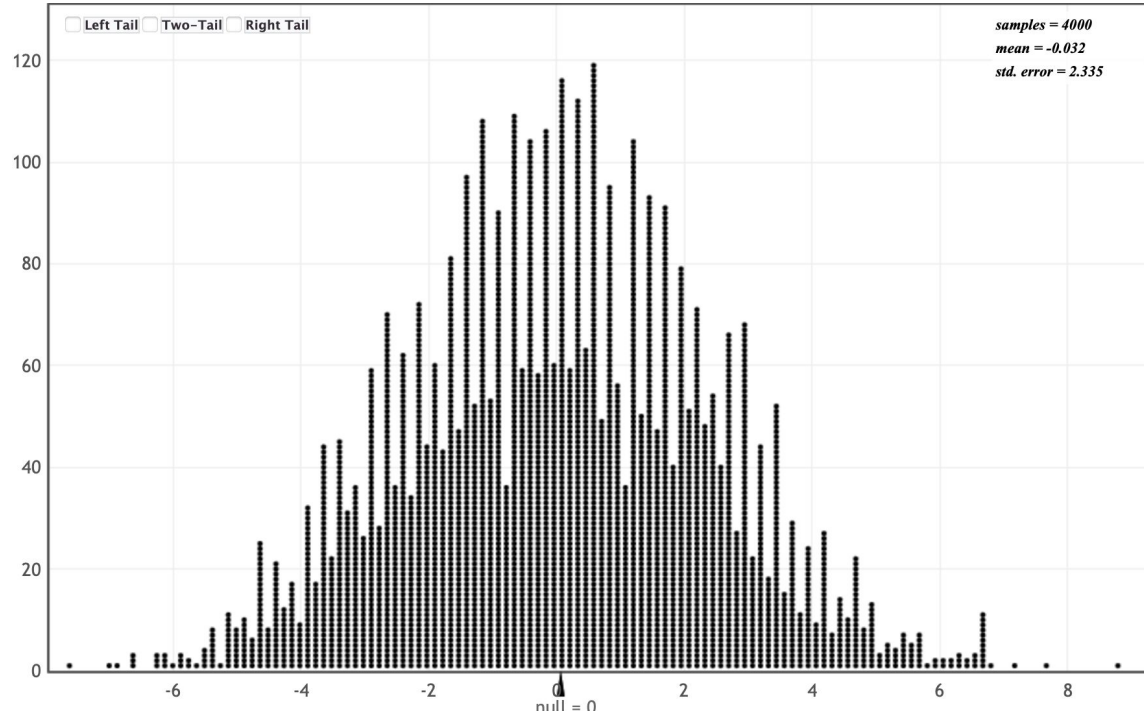


5. What does “95% confidence” mean?

95% confidence means that we are 95% confident that 95% of the time we make a confidence interval, the true population parameter will be within the specified interval / the confidence interval will contain the true population parameter

Group 9 - Randomization Tests

Randomization tests - In Statkey, select the Randomization Test for Difference in Means tool and select “Exercise Hours (Male vs Female).” Generate the randomization test dotplot with a few thousand samples and paste it here.



1. Briefly explain how that dotplot is made. For each dot, what is being calculated? Remember what we did with index cards and the weight loss study.

Each dot represents a simulated difference in means of exercise hours between male and female groups. Under the assumption the null hypothesis is true.

2. What are the null and alternative hypotheses? Use the appropriate symbols.

$H_0: \mu_{\text{male}} - \mu_{\text{female}} = 0$

$H_a: \mu_{\text{male}} - \mu_{\text{female}} > 0$

3. What is the major assumption that underlies this plot?

The null hypothesis is true

4. Where is a randomization test dotplot always centered? Write a sentence explaining why it's centered there.

The randomization test dotplot is centered at 0 because it will always be centered at the null hypothesis value

5. In general, how do you use a sample statistic and the dotplot to draw a conclusion about the difference between the two groups?

We will test for the p-value of Keep going!

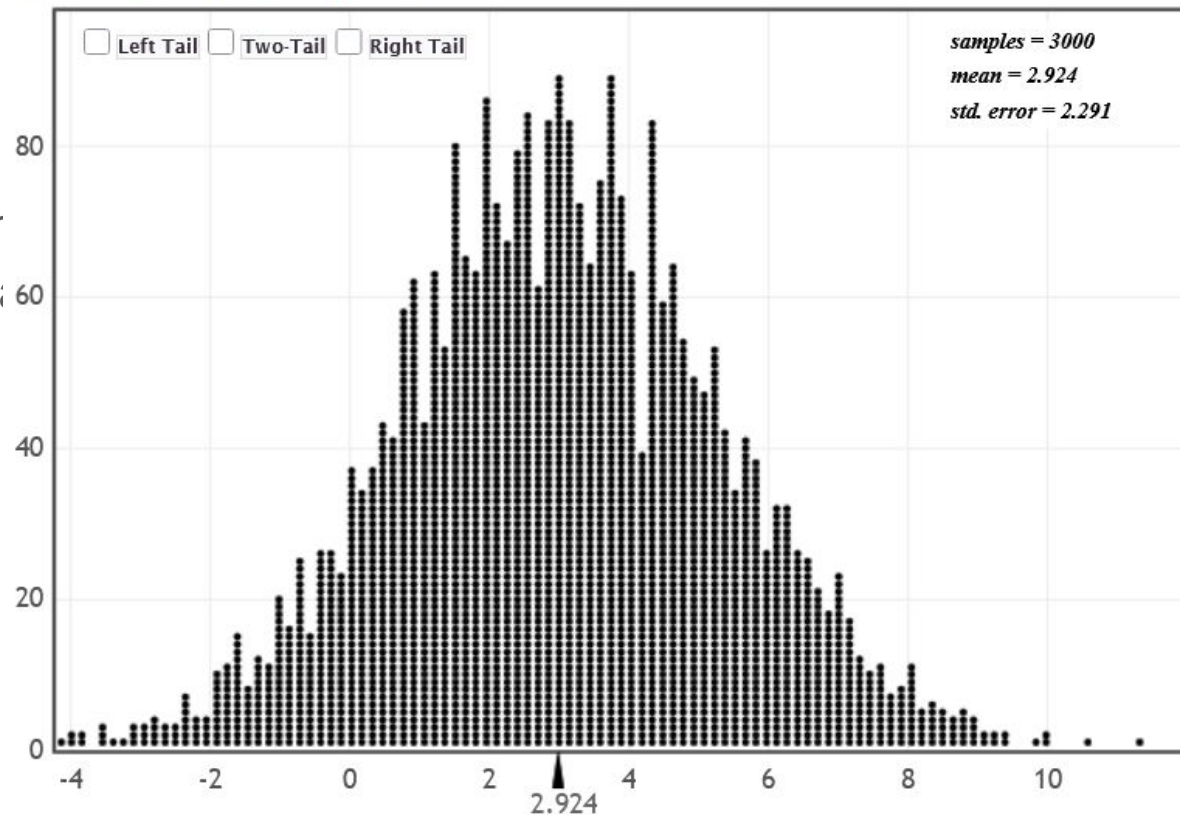
6. What conclusion can you draw from our sample statistic and dotplot? You can use the p-value calculating tool if that would help.

With our sample statistic of 3 hours, we get a p-value of 0.109 which could reasonably be due to random chance, so we will retain the null hypothesis.

Bootstrap sampling distribution

Bootstrap Dotplot of $\bar{x}_1 - \bar{x}_2$

Now select the Confidence Interval for a Difference in Means tool in statkey. Again the exercise hours dataset generate a dotplot.



1. Briefly explain how *this* dotplot is made. For each dot, what is being calculated? How is “bootstrapping” involved?

For this dotplot, every dot is constructed from a random portion of our sample.

2. Where are *these* dotplots always centered? Write a sentence explaining why it makes sense that they are centered there.

The dot plots are always centered at zero. This makes sense because it asserts that there is no difference between groups and the observed difference(if any) is due to random chance.

3. In general, how do you use this dotplot to draw conclusions about a population parameter (or two population parameters, in this case).

We would calculate our standard error to find our desired confidence interval.

4. Follow your procedure to generate the appropriate inference for these variables. What can you conclude? Write your explanation the way we've practiced.

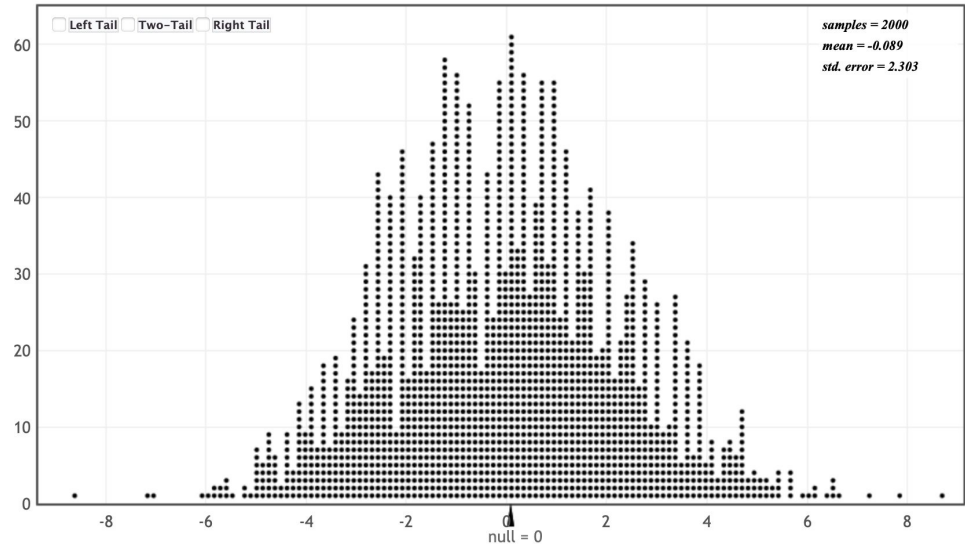
We can conclude that we are 95% confident that that average difference in exercise hours are within the range of -1.390 and 7.950.

5. What does "95% confidence" mean?

95% of the time that we construct the interval in this way, our interval will contain the population parameter.

Group 10 - Randomization Tests

Randomization tests - In Statkey, select the Randomization Test for Difference in Means tool and select “Exercise Hours (Male vs Female).” Generate the randomization test dotplot with a few thousand samples and paste it here.



1. Briefly explain how that dotplot is made. For each dot, what is being calculated? Remember what we did with index cards and the weight loss study.

For each dot on the graph it represents the exercise hours for 20 men and 30 women. Each of the 50 people in the data the hours are then shuffled and redistributed to both the female and males in which the men are given 20 and the women 30.

2. What are the null and alternate hypotheses? Use the appropriate symbols.

$\mu_{\text{male}} - \bar{x}_{\text{female}} = 0$: The gender did not have an effect on the amount of hours worked out

$\mu_{\text{male}} - \bar{x}_{\text{female}} > 0$: The gender did have an effect on the amount of hours worked out

3. What is the major assumption that underlies this plot?

That men will have more workout hours than that of a women

4. Where is a randomization test dotplot always centered? Write a sentence explaining why it's centered there.

A randomization test dotplot is always centered in the middle making a bell shaped graph. It is centered at zero because the graph represents the sample data which is set at zero as a standard.

5. In general, how do you use a sample statistic and the dotplot to draw a conclusion about the difference between the two groups?

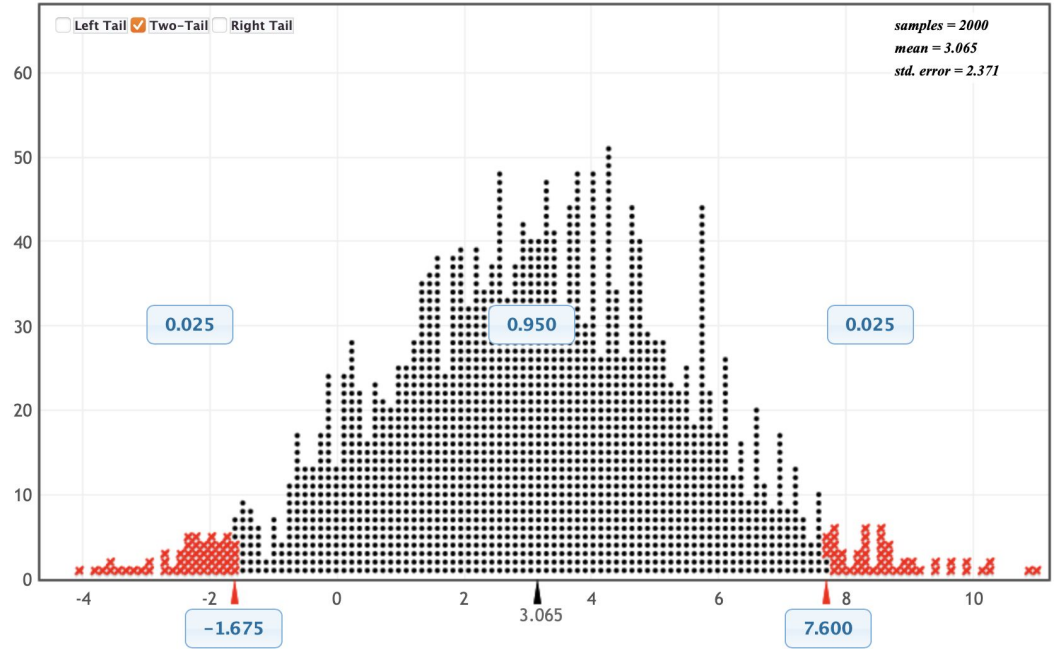
Looking at the differences between shape, center and spread to decide the differences between the 2 **Malec here - what does this mean? We only have one dotplot!**

6. What conclusion can you draw from our sample statistic and dotplot? You can use the p-value calculating tool if that would help.

Our p-value is around .112 meaning that it isn't very unusual for this to happen. It could have happened in chance alone or just complete differences between the groups. We can reject the null

Bootstrap sampling distribution

Now select the Confidence Interval for a Difference in Means tool in statkey. Again find the exercise hours dataset and generate a dotplot.



1. Briefly explain how *this* dotplot is made. For each dot, what is being calculated? How is “bootstrapping” involved?

The graph takes a mean of each randomized hours exercised of 50 people and puts a dot on the graph. For each dot the difference of means of the 50 people who have exercised is being calculated.

2. Where are *these* dotplots always centered? Write a sentence explaining why it makes sense that they are centered there.

They are centered around the sample statistic. It makes sense that its centered around there so we can see the complete differences on each sample

3. In general, how do you use this dotplot to draw conclusions about a population parameter (or two population parameters, in this case).

You look at the shape, center, and spread again allowing you to determine conclusions **Malec - gotta be way more specific!**

4. Follow your procedure to generate the appropriate inference for these variables. What can you conclude? Write your explanation the way we've practiced.

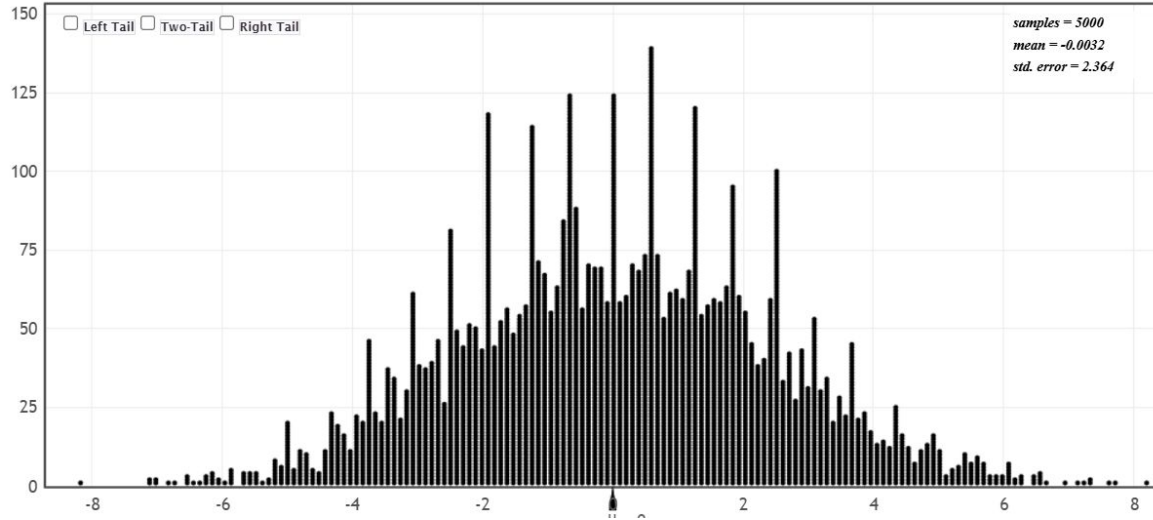
Based on the bootstrapping data we have gotten from the graph we are 95% confident the mean amount of hours that 50 people have exercised is going to be between -1.67 and 7.60.

5. What does "95% confidence" mean?

It means that the difference in means that is calculated shows that we are 95% confident that the number we have calculated is going to be the mean.

Group 11 - Randomization Tests

Randomization tests - In Statkey, select the Randomization Test for Difference in Means tool and select “Exercise Hours (Male vs Female).” Generate the randomization test dotplot with a few thousand samples and paste it here.



1. Briefly explain how that dotplot is made. For each dot, what is being calculated? Remember what we did with index cards and the weight loss study.

For each dot the difference between the means for those in the data set that are female and those in the data set that are male. It then assumes the null hypothesis and tests the null hypothesis thousands of times to see what the likelihood of the alternate hypothesis is, using a confidence interval.

2. What are the null and alternate hypotheses? Use the appropriate symbols.

The null hypothesis is that the data is that the data randomly happened to be this way and there is no outside factor that affected it. ($\mu_1 = \mu_2$)

The alternate hypothesis is that the group that the people are in has an effect on the average difference in the data, this can be calculated with the formula $\bar{x}_1 - \bar{x}_2$

3. What is the major assumption that underlies this plot?

The null hypothesis is true

4. Where is a randomization test dotplot always centered? Write a sentence explaining why it's centered there.

It should always be centered at 0 because the randomly sampled differences in means should always create a bell shaped plot with a center around 0

5. In general, how do you use a sample statistic and the dotplot to draw a conclusion about the difference between the two groups?

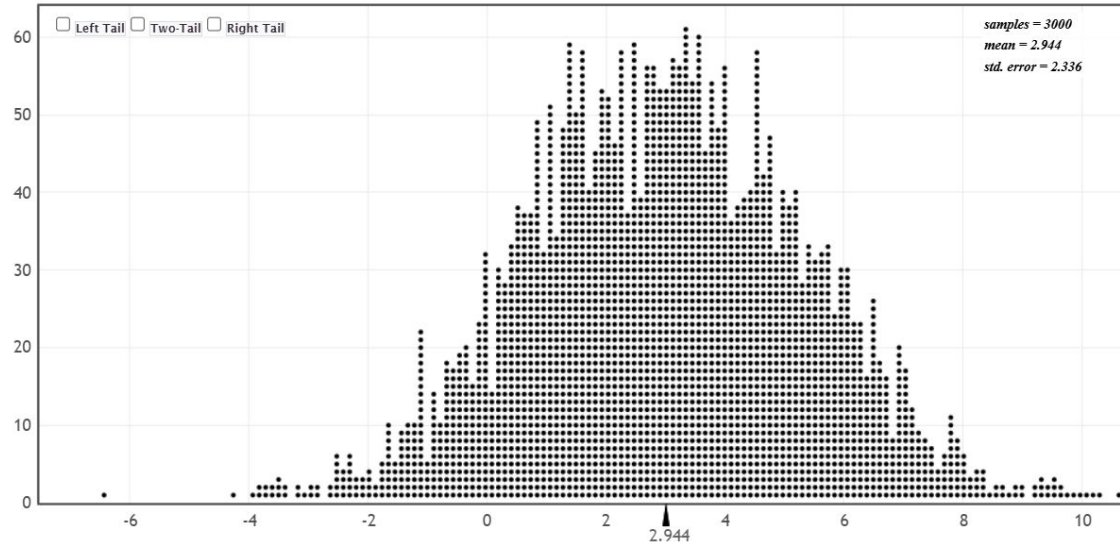
You can use the dot plot and a sample statistic to draw the conclusion that the null hypothesis should be retained

6. What conclusion can you draw from our sample statistic and dotplot? You can use the p-value calculating tool if that would help.

We can conclude that the null hypothesis should be retained due to the difference in the average weekly workout hours being plausible.

Bootstrap sampling distribution

Now select the Confidence Interval for a Difference in Means tool in statkey. Again find the exercise hours dataset and generate a dotplot.



1. Briefly explain how *this* dotplot is made. For each dot, what is being calculated? How is “bootstrapping” involved?

This dot plot is made by taking the difference in means for many randomly generated data sets. It then uses the difference in means to calculate a confidence interval. Then the confidence interval is plotted and used to create the bootstrapping plot.

2. Where are *these* dotplots always centered? Write a sentence explaining why it makes sense that they are centered there.

They are centered around the sample statistic

3. In general, how do you use this dotplot to draw conclusions about a population parameter (or two population parameters, in this case).

You can use the mean value of this plot to understand what the average difference in time working out between men and women is.

4. Follow your procedure to generate the appropriate inference for these variables. What can you conclude? Write your explanation the way we've practiced.

Based on the data we are 95% confident that the difference in average workout hours between men and women will fall between -1.492 and 7.475 hours

5. What does "95% confidence" mean?

95% of the time the sample will contain a range including the true population parameter